

WASP-39b

R-1841

Benchmark run · workflow W8-benchmark-deep · record deep-bench_WASP-39_250607 · created 2026-08-03T06:58:13+00:00

PRACTICE / VALIDATION RUN — not submitted to any science database

Results

Mid-Transit Time (Tmid)	2460833.814 +/- 0.015 BJD_TDB
Ratio of Planet to Stellar Radius (Rp/R*)	0.191 +/- 0.055
Transit depth (Rp/Rs)^2	3.64 +/- 2.09 [%]
Orbital Inclination (inc)	88.02 +/- 1.93
Airmass coefficient 1 (a1)	0.909 +/- 0.046
Airmass coefficient 2 (a2)	0.0316 +/- 0.0093
Scatter in the residuals of the lightcurve fit is	5.12 %
Best Comparison Star	#2 - [572, 464]
Optimal Aperture	2.34
Optimal Annulus	15.33
Transit Duration (day)	0.101 +/- 0.025

Accuracy vs published ephemeris (ExoClock/NEA)

Rp/R■ fit vs published	0.191 ± 0.055 vs 0.1457 (deviation 0.82σ)
Duration fit vs published	0.101 d vs 0.12 d (deviation 0.76σ)
Mid-time O–C	-7.3 min
Depth SNR	3.5
Residual scatter	5.12 %

Light curve

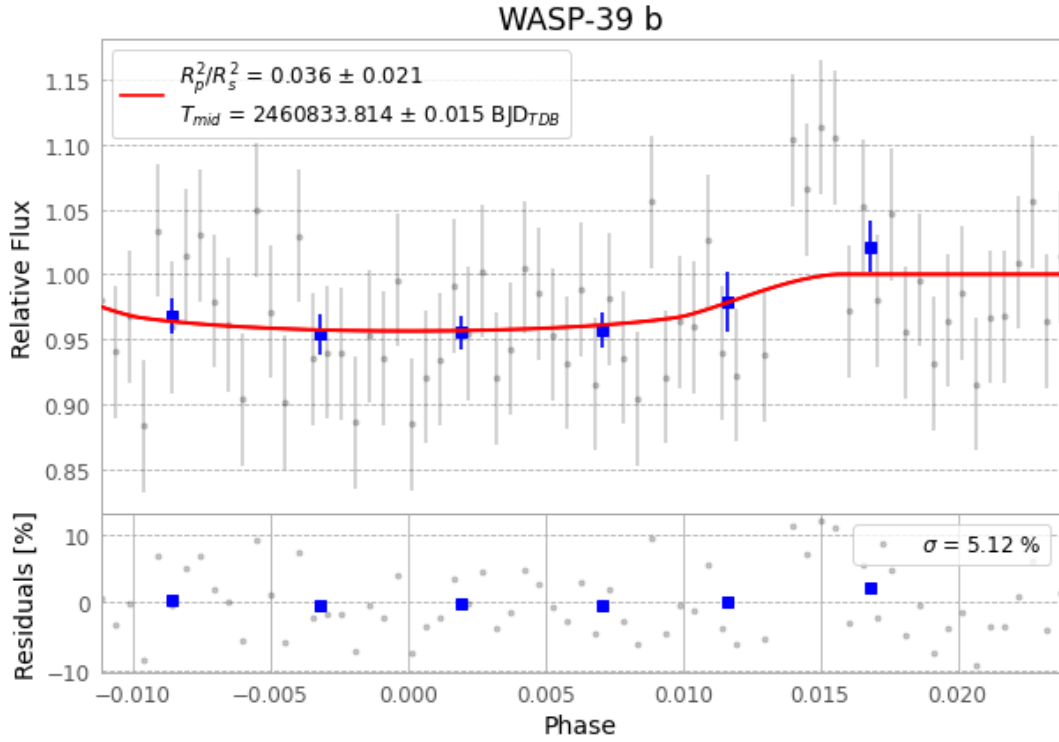


Figure 1 — FinalLightCurve_WASP-39 b_2025-06-06.png (SHA-256 8baa774c2b494bcf...)

Method & software

EXOTIC 4.3.1 aperture photometry with nested-sampling transit fit (ultranest 3.6.5); priors from the NASA Exoplanet Archive. Camera CCD, binning 1x1, filter CV, target pixel [300, 246], 3 comparison star(s). Exact configuration: parameters.inits.json in record.json (SHA-256 5ddd30efca2d7bf3...).

exotic 4.3.1 · astropy 6.1.7 · photutils 2.0.2 · numpy 1.26.4 · scipy 1.14.1 · twirl 0.5.1 · astroquery 0.4.11 · ultranest 3.6.5 · python 3.12.13 · pipeline_commit ba3ef8b

Data provenance

69 FITS frames (45 MB), WASP-39250607061815.FITS → WASP-39250607094217.FITS. Every input frame is SHA-256-fingerprinted in record.json; raw frames available on request and verifiable against that manifest. Artifacts: bench-WASP-39-250607.log (c707806dbc9d...), FinalLightCurve_WASP-39 b_2025-06-06.png (8baa774c2b49...), FinalLightCurve_WASP-39 b_2025-06-06.csv (1c3e90749bfd...)

Compute resources

Wall time 145.2 s · peak memory None MB · host mac-mini-m1-8gb (macOS-26.4-arm64-arm-64bit)

Observer: — · AAVSO obscode ABGA · Automated pipeline with human review — nothing is submitted without a human approving this specific result. Data license CC-BY-4.0. Generated 2026-08-03 06:58Z.